

Book 6: Poultry Health, Biosecurity & Disease Management

A Standard Operating Guide to Disease Prevention, Vaccination Protocols & Flock Diagnostics

PoultryWise Operational Manuals • Veterinary Health & Biosecurity Standard

TABLE OF CONTENTS

Chapter 1: Foundations of Structural & Operational Biosecurity

Chapter 2: Farm Sanitation, Washing & Disinfection Standards

Chapter 3: Vaccination Strategy & Cold Chain Management

Chapter 4: Viral Diseases: Newcastle, Gumboro & IB

Chapter 5: Avian Influenza Preparedness & Response

Chapter 6: Bacterial Diseases & Antimicrobial Stewardship

Chapter 7: Parasitic & Coccidiosis Management

Chapter 8: Post-Mortem Diagnostics & Lesion Identification

Chapter 9: Serological Monitoring & Immunity Auditing

Chapter 10: Emergency Mortality Disposal & Outbreak Control

CHAPTER 1: FOUNDATIONS OF STRUCTURAL & OPERATIONAL BIOSECURITY

Biosecurity is the most cost-effective disease control measure in commercial poultry production. It encompasses physical barriers and operational protocols designed to prevent pathogen entry and spread.

• Three Zones of Biosecurity:

- *Clean Zone:* Inside the poultry house; accessible only after shower-in/shower-out and changing into farm-dedicated coveralls and boots.
- *Buffer Zone:* Farm perimeter containing feed bins, egg stores, and equipment sheds. Vehicle wheels must pass through a 10% disinfectant bath.
- *Dirty Zone:* Public roads, visitor parking, and mortality pickup points outside the farm fence.

- **All-In / All-Out (AIAO) Policy:** Depopulate entire sites simultaneously to allow complete cleaning and break disease cycles before stocking the next flock.

CHAPTER 2: FARM SANITATION, WASHING & DISINFECTION STANDARDS

Proper terminal hygiene between flocks removes pathogen loads and organic debris from housing equipment and surfaces.

Standard 6-Step Terminal Cleaning Protocol

1. **Dry Cleaning & Litter Removal:** Remove all litter, feed, and organic dust completely from the house.
2. **Pre-Soaking:** Spray low-pressure water mixed with a neutral detergent to soften caked organic matter.
3. **High-Pressure Washing:** Power wash all structural surfaces, fans, and curtains with hot water (160°F).
4. **Drying:** Allow surfaces to dry completely before disinfectant application (water dilutes chemical active ingredients).
5. **Disinfection:** Apply broad-spectrum disinfectant (Glutaraldehyde + Quaternary Ammonium Compound) using a power sprayer.
6. **Fumigation & Downtime:** Seal the house for formal/aerosol fogging; maintain a minimum 14-day downtime before chick placement.

CHAPTER 3: VACCINATION STRATEGY & COLD CHAIN MANAGEMENT

Vaccines train the bird's immune system to recognize and fight viral and bacterial pathogens. Efficacy depends entirely on cold-chain integrity and proper administration.

Target Age	Disease Target	Vaccine Type	Administration Route
Day 0 (Hatchery)	Marek's Disease, HVT / IBD	Live / Recombinant (HVT)	Subcutaneous / In-Ovo
Day 7 – 10	Newcastle Disease (ND) + IB	Live (B1 / LaSota + Ma5)	Eye drop / Fine Spray
Day 14 – 18	Infectious Bursal Disease (Gumboro)	Live Intermediate Plus	Drinking Water
Week 8 & Week 14	ND + IB + EDS + Coryza (Layers)	Inactivated Oil Adjuvant	Intramuscular / Subcut

CHAPTER 4: VIRAL DISEASES: NEWCASTLE, GUMBORO & IB

Viral infections cause major mortality, egg drops, and immunosuppression in commercial layers and broilers.

- **Newcastle Disease (ND):** Caused by Avian Paramyxovirus-1. Symptoms include torticollis (twisted neck), green diarrhea, respiratory distress, and up to 100% mortality in velogenic strains.
- **Infectious Bursal Disease (IBD / Gumboro):** Targets the Bursa of Fabricius in young birds (3–6 weeks). Causes severe immunosuppression, hemorrhages in thigh muscles, and enlarged, gelatinous bursae.
- **Infectious Bronchitis (IB):** Coronavirus targeting respiratory and reproductive tracts. Causes gasping, misshapen eggs with thin shells, and permanent kidney damage (nephropathogenic strains).

CHAPTER 5: AVIAN INFLUENZA PREPAREDNESS & RESPONSE

Highly Pathogenic Avian Influenza (HPAI - H5N1, H5N8) is a global threat capable of wiping out whole flocks within hours.

- **Clinical Signs:** Sudden spike in mortality, cyanosis (blue discoloration) of combs/wattles, facial edema, leg shank hemorrhages, and severe drop in feed and water intake.

- **Wild Bird Exclusion:** Cover all ventilation openings with wild bird netting (< 25mm mesh). Prevent waterfowl from accessing open farm water reservoirs.
- **Action Protocol on Suspicion:** Immediately quarantine the farm, halt bird and feed movements, and notify official state veterinary authorities.

CHAPTER 6: BACTERIAL DISEASES & ANTIMICROBIAL STEWARDSHIP

Bacterial pathogens frequently act as secondary invaders following viral exposure or environmental stress.

Disease	Causative Agent	Key Lesions / Symptoms	Control & Prevention
Colibacillosis	<i>Escherichia coli</i>	Fibrinous pericarditis, perihepatitis, airsacculitis	Water sanitation, ventilation, probiotic support
Infectious Coryza	<i>Avibacterium paragallinarum</i>	Facial swelling, foul-smelling nasal discharge	Bacterin vaccination, strict multi-age separation
Fowl Cholera	<i>Pasteurella multocida</i>	Sudden death, swollen wattles, consolidated lungs	Rodent control, flock sanitation, targeted antibiotics

CHAPTER 7: PARASITIC & COCCIDIOSIS MANAGEMENT

Coccidiosis, caused by *Eimeria* protozoa, damages the intestinal mucosa, leading to poor Feed Conversion Ratio (FCR) and secondary necrotic enteritis.

- **Key Eimeria Species:**
 - *Eimeria tenella*: Affects the ceca; causes severe bloody droppings and cecal core accumulation.
 - *Eimeria acervulina*: Affects upper intestine; causes white transverse ladder-like lesions.
 - *Eimeria necatrix*: Affects mid-intestine; causes severe hemorrhage and ballooning.
- **Control Strategies:** Shuttle and rotation programs using ionophores (Monensin, Salinomycin) and synthetic chemical anticoccidials (Diclazuril, Toltrazuril) or live coccidiosis vaccination.

CHAPTER 8: POST-MORTEM DIAGNOSTICS & LESION IDENTIFICATION

Field necropsy provides rapid diagnostic feedback to guide immediate flock treatment decisions.

Systematic Necropsy Sequence

Always perform necropsies systematically to avoid missing critical diagnostic clues:

- **External Inspection:** Assess comb color, eyes, nostrils, vent, plumage condition, and dehydration status.
- **Abdominal Cavity:** Examine liver, heart, spleen, and abdominal fat pad for fibrinous exudates or hemorrhages.
- **Digestive Tract:** Open proventriculus, gizzard, small intestine, and ceca to inspect mucosa for ulcers, blood, or parasites.
- **Immune & Respiratory Organs:** Check size and texture of the Bursa of Fabricius, thymus, trachea, and lungs.

CHAPTER 9: SEROLOGICAL MONITORING & IMMUNITY AUDITING

Routine blood sampling (ELISA testing) verifies vaccine uptake, maternal antibody decay, and field challenge exposure.

- **Baseline Titer Tracking:** Collect 20 serum samples per flock at key production milestones (e.g., 6, 12, 18, 24 weeks).
- **Interpreting Results:** High Geometric Mean Titers (GMT) with a low Coefficient of Variation (< 30%) indicate uniform and effective vaccine coverage. High titers accompanied by high CV values signal uneven vaccination or field challenge.

CHAPTER 10: EMERGENCY MORTALITY DISPOSAL & OUTBREAK CONTROL

Safe disposal of dead birds is critical to prevent environmental contamination and pathogen spread across farms.

- **Composting:** Standard on-farm disposal method using alternating layers of carbon (sawdust/straw), manure, and mortality (1:2 ratio). Maintains core temperatures exceeding 55°C - 60°C to kill pathogens.
- **Incineration:** High-temperature burning ($> 850^{\circ}\text{C}$) ideal for high-virulence outbreaks (Avian Influenza, velogenic ND).
- **Deep Burial:** Permitted under specific regulatory oversight; pits must be lined with quicklime and located away from groundwater tables.

End of Book 6 — PoultryWise Knowledge Series • Poultry Health, Biosecurity & Disease Management